

Time Light: An Interface for Comparing National Treasure Murals Across Time

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(a)



(b)



(c)

Figure 1: (a) One of the murals from the Takamatsuzuka Kofun, Asuka Bijin, showing severe damage. (b) The reconstructed murals based on research and investigation. (c) Observing and comparing the damaged and reconstructed murals simultaneously using Time Light.

Abstract

Time Light is an innovative interface developed as part of a Mixed Reality (MR) system designed to deepen the understanding of the renowned Takamatsuzuka Kofun's murals. This interface allows users to compare and observe both the severely deteriorated murals, affected by looting and plaster degradation at the time of discovery, and the estimated reconstructed images that illustrate how the murals might have appeared when originally created. Although Time Light was developed to enhance the understanding of the Takamatsuzuka Kofun, it can be applied to any murals with both discovered and reconstructed images. By using Time Light, users can experience a unique perspective on the historical and cultural significance of these national treasures.

CCS Concepts

• **Human-centered computing** → **Systems and tools for interaction design**; • **Applied computing** → *Fine arts*; Interactive learning environments.

Keywords

interaction design, mixed reality, virtual reality, takamatsuzuka kofun, mural painting, cultural property, digital content

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1 Introduction

Famous paintings and murals, along with other two-dimensional artworks, are often carefully preserved in museums and art galleries. However, many paintings have suffered significant damage due to fires, looting, and improper storage. Although these artworks are frequently restored or repainted by experts, and their digital versions are widely accessible and can be viewed in various forms, such as zooming and rotating, allowing more people to appreciate them and understand their historical context and stories, there remains a challenge. Even when both the damaged and restored images are publicly available, there are many outstanding interactive systems designed to enhance the understanding of the mural restoration process[Fu et al. 2020]. However, it is still difficult for viewers to effectively compare and understand the efforts involved in their restoration.

For example, the focus of this study is the Takamatsuzuka Kofun [J. Edward Kidder 1972]. This ancient tomb, located in Asuka-mura, Nara, Japan, dates back to the late 7th to early 8th century. It is renowned for its vividly colored and diverse murals, including depictions of human figures, the Four Gods, celestial maps, and constellations. However, due to looting and plaster erosion, the condition of these murals had significantly deteriorated by the time they were discovered[Li et al. 2012] (Figure 1(a)). Researchers from Kansai University created reconstructed images based on their studies, as shown in Figure 1(b)[Museum [n. d.]].

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In this study, we have developed an MR system that places the digitized Takamatsuzuka Kofun in an MR environment, allowing viewers to intuitively observe the Kofun's dimensions and murals within the surrounding context. Additionally, we have developed a user interface system called "Time Light," which enables users to simultaneously view and compare the state of the murals at the time of their discovery, half a century ago, with their estimated appearance when originally created 1300 years ago. This paper will detail the usage, implementation, and user feedback of "Time Light" (Figure 1(c)).

2 Experience Design

2.1 Experiential MR System

We have developed a program for use with the Meta Quest 3. In this program, users can freely walk around in MR and pass directly through the walls of the Kofun to view the murals inside, increasing the accessibility of the Kofun, which would otherwise be inaccessible due to its narrow space. This system utilizes the hand-tracking functionality of the Meta Quest 3, allowing users' real-world hand movements to be mapped into the MR world and intuitively use the Time Light feature.

2.2 Time Light

The use of the Time Light is illustrated in Figure 2(a). The user's right index finger in the virtual space emits a beam like a flashlight. When this beam illuminates the Kofun, the deteriorated murals in the illuminated area transform into the reconstructed image.

The implementation principle is shown in Figure 2(b). First, (1) the restored image is overlaid on (2) the damaged image. Then, a script adjusts the visibility of the restored image, making it more visible where the light shines. In this experiment, we used Unity's Shader scripts to achieve this effect.

Considering Distance Attenuation and the Spotlight Effect, we set the relationship between transparency (α) and light intensity (I) as shown in the following equation. Where d is the **distance between the object and the light source**, θ is the **spotlight's cone angle**, r is the **range of the light source**, and A is the **cosine of the angle between the light direction and the direction to the object**.

$$I = 2.0 \times \frac{1}{0.1 + d^2} \times \text{smoothstep} \left(\cos \left(\frac{\theta}{2} \right), \cos \left(\frac{\theta}{2} \right) + 0.05, A \right)$$

$$\alpha = \text{saturate}(I \times r)$$

3 Exhibition and User Feedback

We exhibited the MR system equipped with the Time Light at Kansai University from August 3-4, 2024, and collected 58 user questionnaires. The 5-point evaluation results regarding whether the Time Light was successfully used and the usability experience are shown in Figure 3 (the higher the number, the better the feedback). As shown in Figure 3(a), 82.8% of participants rated the successful use of the Time Light as 4 (good) or 5 (very good). Figure 3(b) shows that 94.8% of participants rated the usability experience of the Time Light as 4 (good) or 5 (very good). Additionally, many free-text

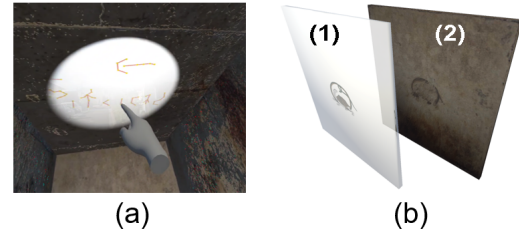


Figure 2: (a) The Time Light shows the reconstructed image where it shines, while the unlit parts remain as they were found, deteriorated. (b) The reconstructed image overlays the murals' discovered state.

comments mentioned that "using the Time Light allowed me to better compare the murals, making it a very interesting experience."

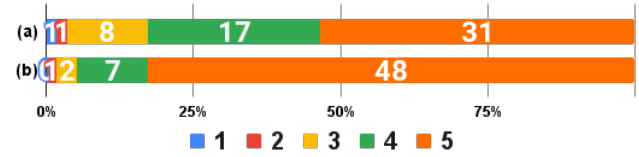


Figure 3: Research Results of the Time Light. (a) Whether it was successfully used. (b) Whether the experience was satisfactory.

4 Conclusion

In this study, we developed a user interaction system called Time Light and integrated it into an MR system to enhance the comparison and observation of murals in the Takamatsuzuka Kofun. During the exhibition, the system received positive feedback. These results indicate that Time Light effectively improves the viewing experience of damaged artworks and their reconstructed images. However, a limitation of this study is that it requires pre-made reconstructed images for comparison. Therefore, future research will explore the possibility of the system automatically generating reconstructed images for the illuminated areas.

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